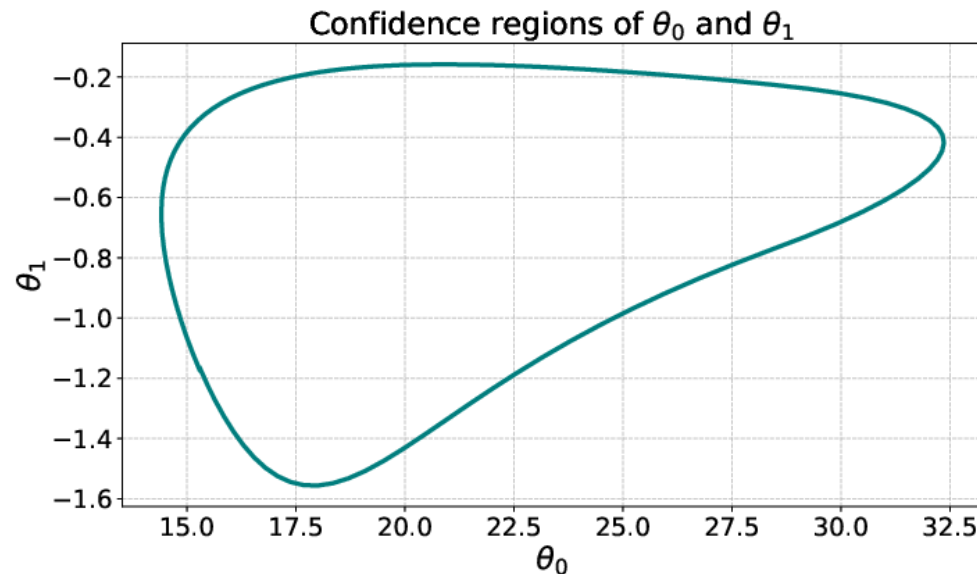


Prediction Intervals and Confidence Regions for Symbolic Regression Models based on Likelihood Profiles

F. Olivetti de Franca, Federal University of ABC

G. Kronberger, University of Applied Sciences Upper Austria (FH OÖ)



Uncertainty in regression analysis

Multiple sources of uncertainty:

- Uncertainty of measurements
- Limited capabilities of observation / sampling
- Uncertainty from choice of model

Often aggregated in an additive error term ϵ as in $y = f(X, \theta) + \epsilon$

Importance of uncertainty quantification

Decision processes may be affected by knowing that the prediction is for 85mm of rain or a high probably interval of [75mm, 110mm].

Point predictions may hide the probabilities for catastrophic or extremely costly events.

Genetic programming for SR traditions

Common in GP/SR papers:

- Machine learning loss functions: Mean-of-squared errors, R^2 , ...
- Least-squares optimization of parameters (or random mutations, ephemeral random constants)
- Ignore data uncertainty
- Ignore parameter uncertainties
- Ignore expression sampling uncertainty

Unknown causes:

- Unawareness of improved methods?
- Concerns about computational complexity?

Bayesian estimation and inference

Assuming a single regression function f_i :

$$P(\theta_i; \mathbf{y}, X, f_i) = \frac{1}{P(D)} P(\mathbf{y}; X, \theta_i, f_i) P(\theta_i; f_i)$$

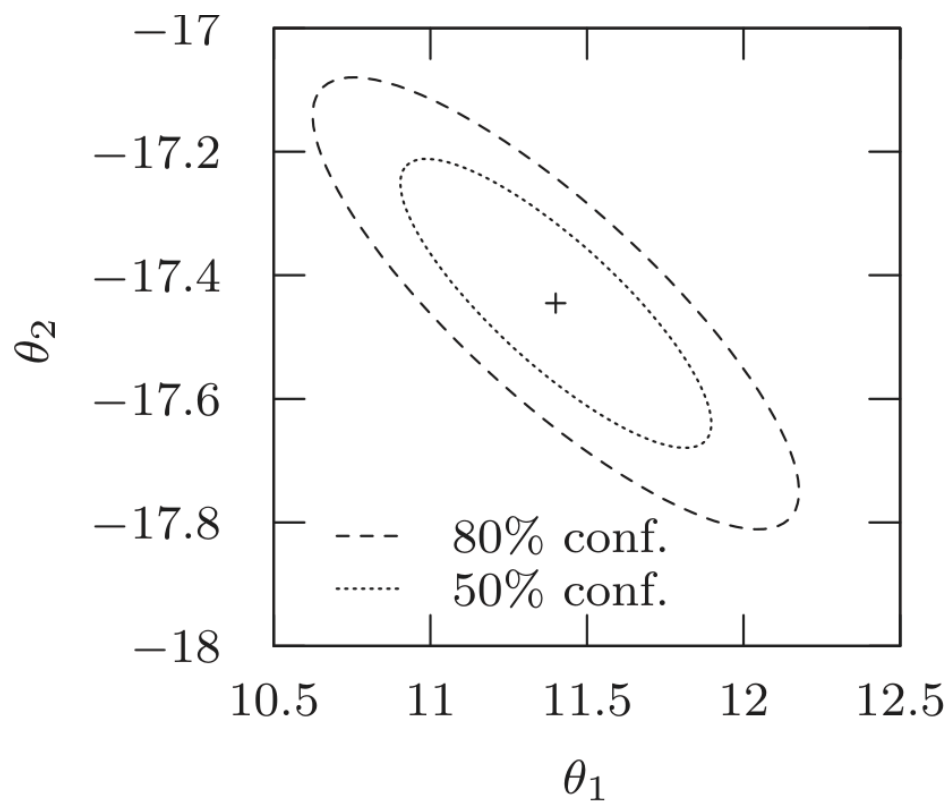
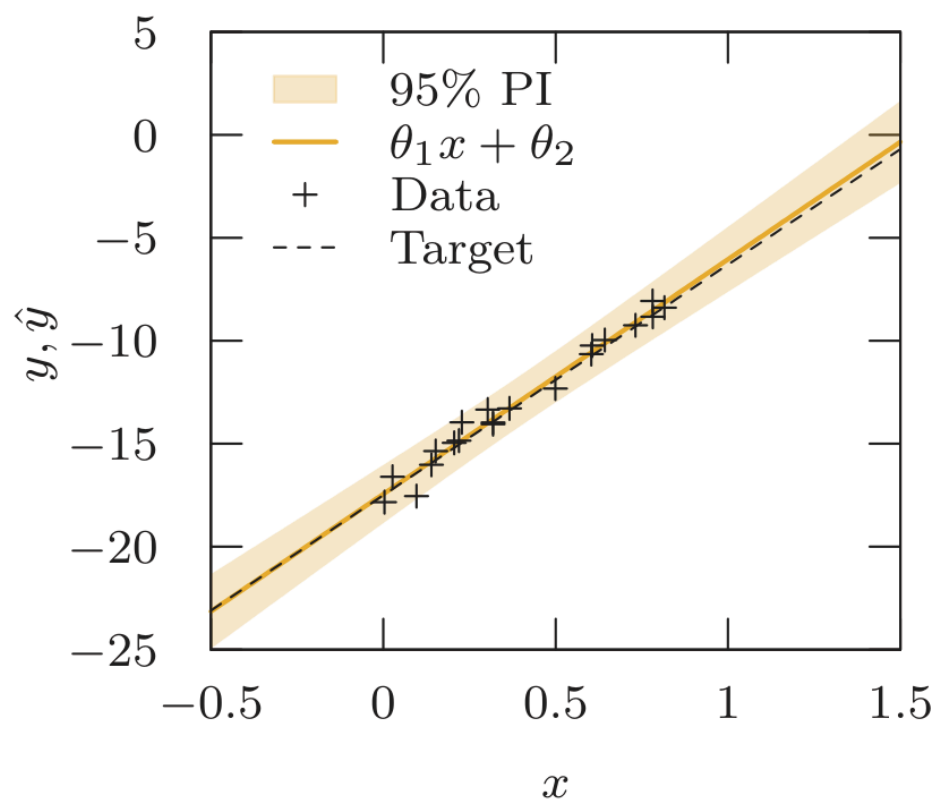
- The posterior distribution over parameters θ_i captures the uncertainty of parameters.
- Probability mass from the prior is moved based on likelihood of observing data under the model
- Uncertainty of data is captured in the likelihood.

Sampling from the posterior is usually no problem with today's hardware and algorithms (exceptions: multi-modal f_i , partial functions, ...).

Nevertheless, posterior approximations can be more efficient.

Bayesian estimation and inference

Predictive distributions for y^* at x^* using the function: $y^* = f_i(x^*, \theta_i)$ on top of the posterior for θ_i .



Simplifications

In the following, we assume:

- A fixed dataset D
- Uniform prior $P(\theta_i; f_i)$

$\Rightarrow P(D)$ and $P(\theta_i, f_i)$ factors cancel in the ratio of posterior probabilities:

$$\frac{P(\hat{\theta}_i; y, x, f_i)}{P(\theta_i; y, x, f_i)}.$$

Only the likelihood ratio remains.

Likelihood-based uncertainty intervals

We assume $\hat{\theta}$ is the maximum likelihood estimate (MLE).

Simplifying notation and taking the log of likelihood ratios:

$$\log \frac{L(\hat{\theta})}{L(\theta)} = \log L(\hat{\theta}) - \log L(\theta)$$

Using a credibility threshold c the set of credible θ values is:

$$\{\theta \mid \log L(\hat{\theta}) - \log L(\theta) > c\}$$

In certain settings, c can be set based on frequentist considerations to get a confidence interval to e.g. capture 95 % of probability mass.

Multiple different approximations for this set...

Laplace approximation

2nd-order Taylor expansion of the log-likelihood ($\ell(\theta) = \log L(\theta)$) around the MLE $\hat{\theta}$

$$\ell(\hat{\theta} + \delta) = \ell(\hat{\theta}) + \nabla \ell(\hat{\theta}) \cdot \delta + \frac{1}{2} \delta^T H(\hat{\theta}) \delta$$

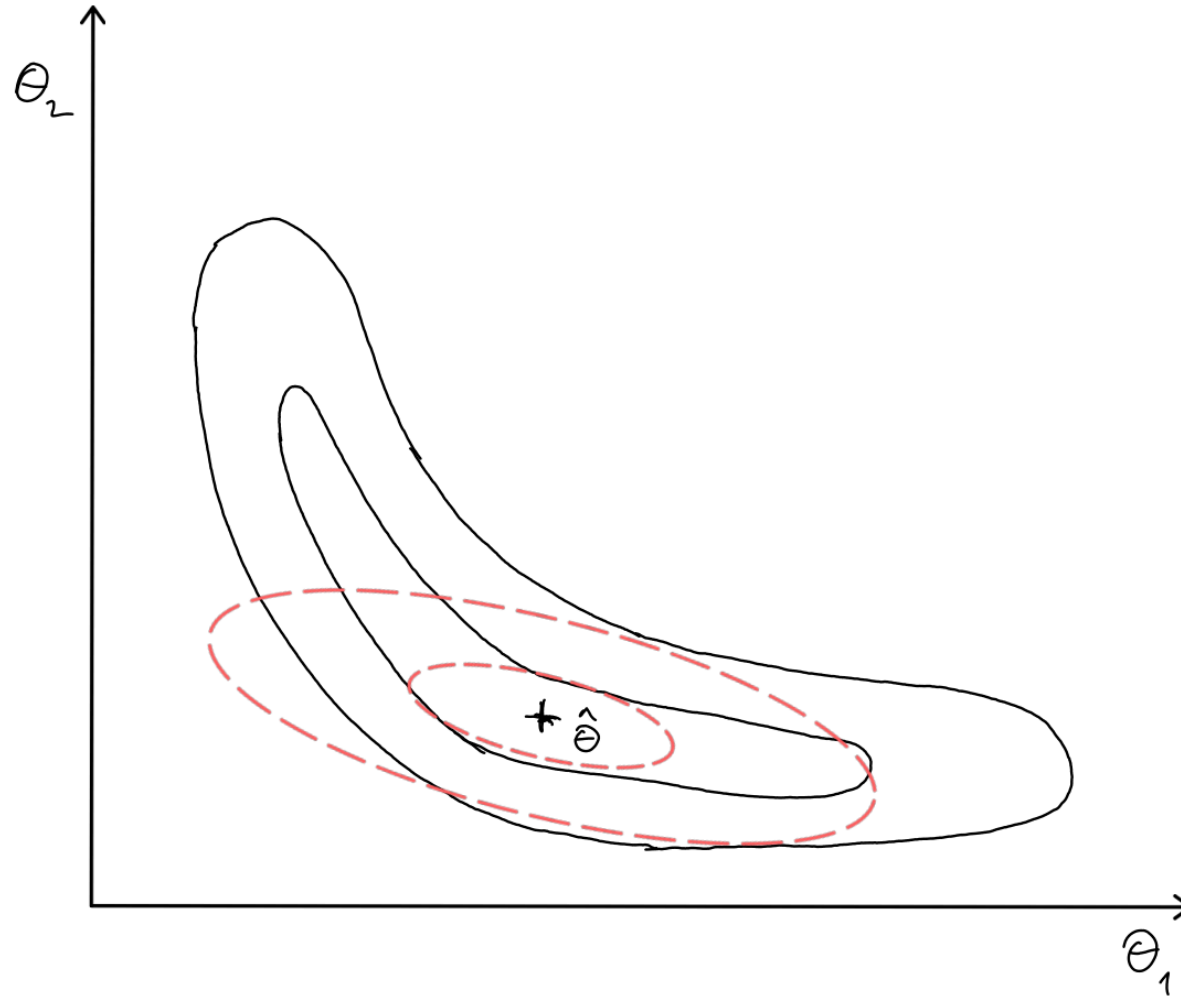
where $H(\hat{\theta})$ is the Hessian of the log-likelihood evaluated at $\hat{\theta}$.

For large number of observations n , the uncertainty in $\hat{\theta}$ becomes small, and the Taylor expansion is close to the likelihood in the relevant region.

Different names depending on circumstances: linear approximation, delta method.

The resulting distribution is a multivariate Gaussian with mean $\hat{\theta}$ and covariance matrix $-H(\hat{\theta})^{-1}$.

Laplace approximation can produce bad results



Profile likelihood approximation

The profile likelihood is also a unimodal approximation around the MLE, but takes the nonlinearity of parameters into consideration.

The profile likelihood considers only one parameter θ_i and uses optimal values for all other parameters $\theta_{j \neq i}$.

$$L_{\text{prof}}(\theta_i) = \max_{\theta_{j \neq i}} L(\theta)$$

Computing the profile likelihood function (algorithm 1)

For computing values of the profile likelihood function, it is recommended to start at the MLE and make small steps for θ_i , optimizing $\theta_{j \neq i}$ in each step.

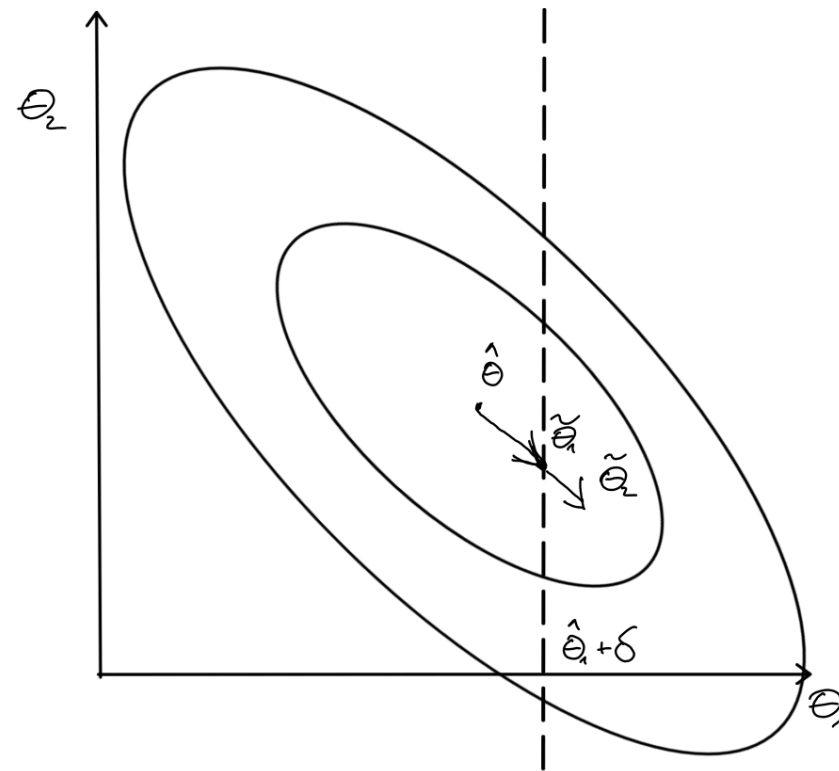
Algorithm 1 Likelihood profile algorithm as described in [1],
 $\mathbb{1}_i$ is a vector with 1 at position i and 0 otherwise.

```
1: function PROFILE( $i, \hat{\theta}, f, J, \delta$ )  
2:    $P \leftarrow \{\}, t \leftarrow 1, \text{it} \leftarrow 1, \tilde{\theta} \leftarrow \hat{\theta}$   
3:   while  $\text{it} \leq 30$  and  $|\tau(\tilde{\theta})| \leq \tau_{max}$  do  
4:      $\tilde{\theta} \leftarrow \tilde{\theta} + \mathbb{1}_i \delta t$   
5:      $\tilde{\theta} \leftarrow \text{optimize}(\tilde{\theta}, f, \tilde{J})$   
6:      $t \leftarrow \text{update}(t, \tau(\tilde{\theta}))$  ▷ Eq. 12  
7:      $P \leftarrow P \cup \{(\tilde{\theta}, \tau(\tilde{\theta}))\}$   
8:      $\text{it} \leftarrow \text{it} + 1$   
9:   end while  
10:  return  $P$   
11: end function
```

Optimization is cheap, because the optimized values are expected to be close to the previous value.

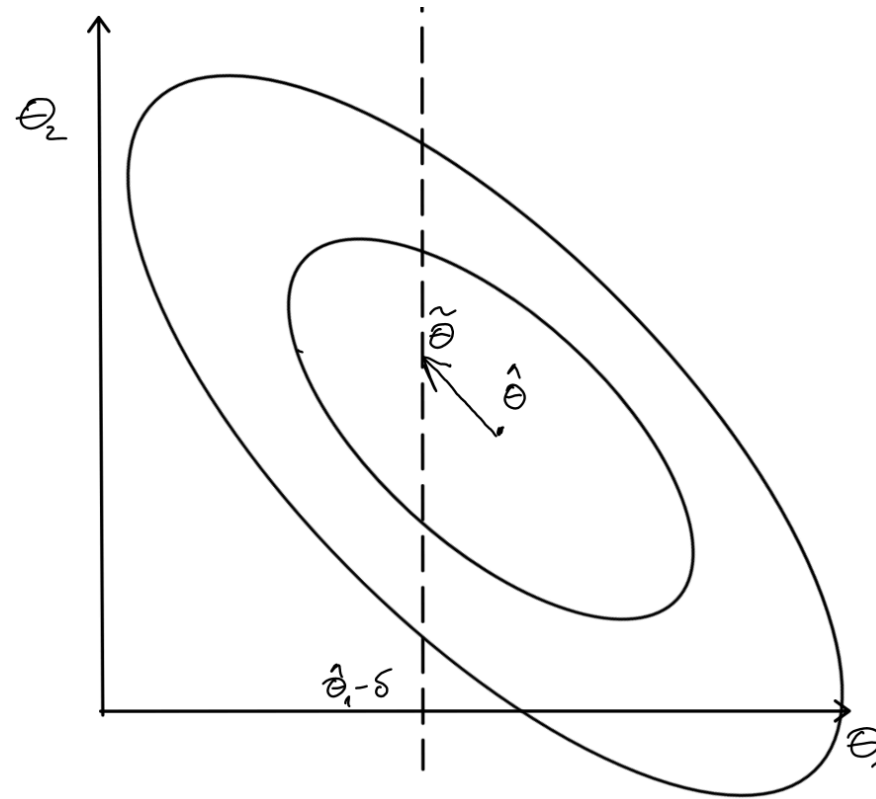
Profile likelihood example

Profile θ_1 in positive direction



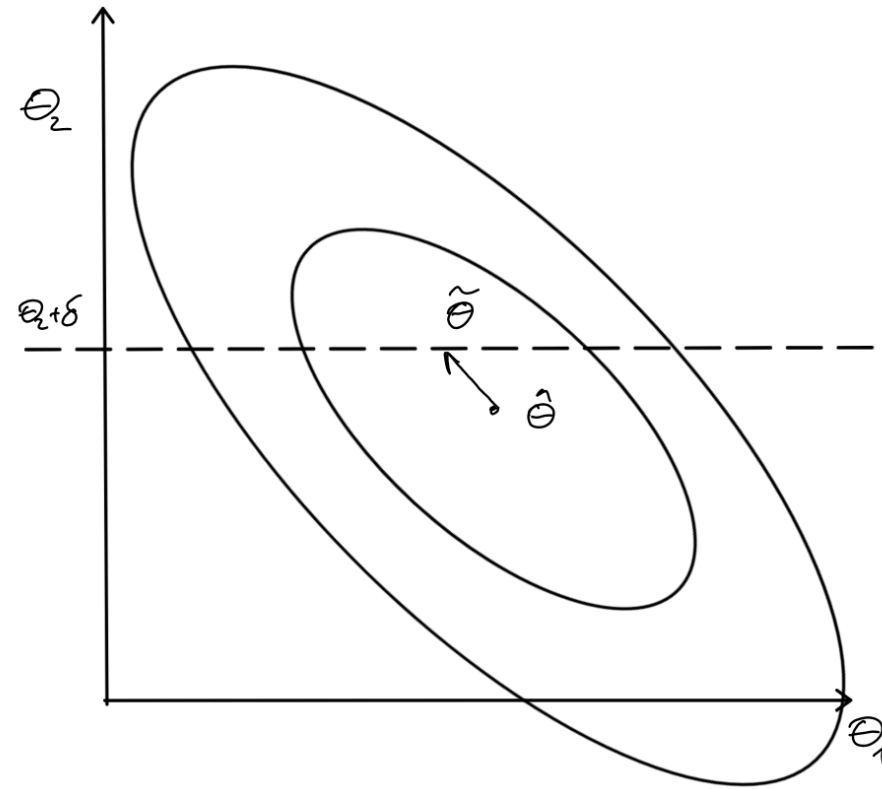
Profile likelihood example

Profile θ_1 in negative direction



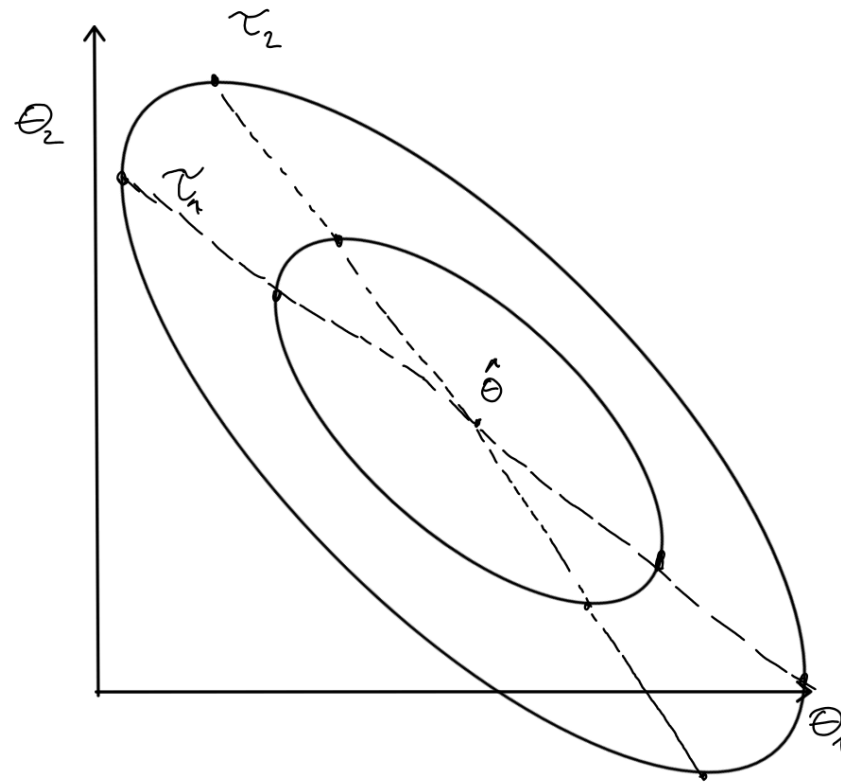
Profile likelihood example

Profile θ_2 in positive direction



Profile likelihood example

Both profiles. Along the path for τ_1 : $\frac{\partial}{\partial \theta_2} \ell(\theta) = 0$ and vice versa.



Likelihood profiling as ODE (algorithm 2)

Instead of repeated optimisation, solve a system of ordinary differential equations (ODE) using the parameter vector as state vector.

The system is defined as:

$$\begin{pmatrix} \text{FI}(\theta) & \mathbf{g} \\ \mathbf{g} & 0 \end{pmatrix} \begin{pmatrix} \dot{\theta} \\ \mu \end{pmatrix} = \begin{pmatrix} -\gamma \dot{\ell}(\theta) \\ 1 \end{pmatrix}$$

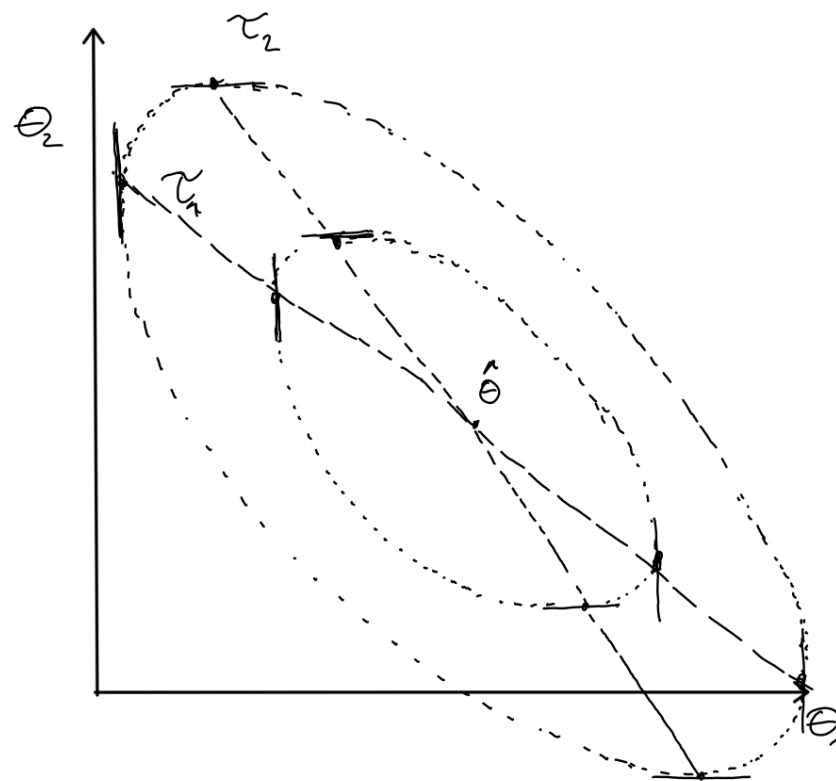
$\text{FI}(\theta)$ is the Fisher information matrix evaluated at θ
 $\mathbf{g}$ has a single 1 at position i .

The ODE has to be solved numerically for each parameter and in both directions with initial value $\hat{\theta}$.

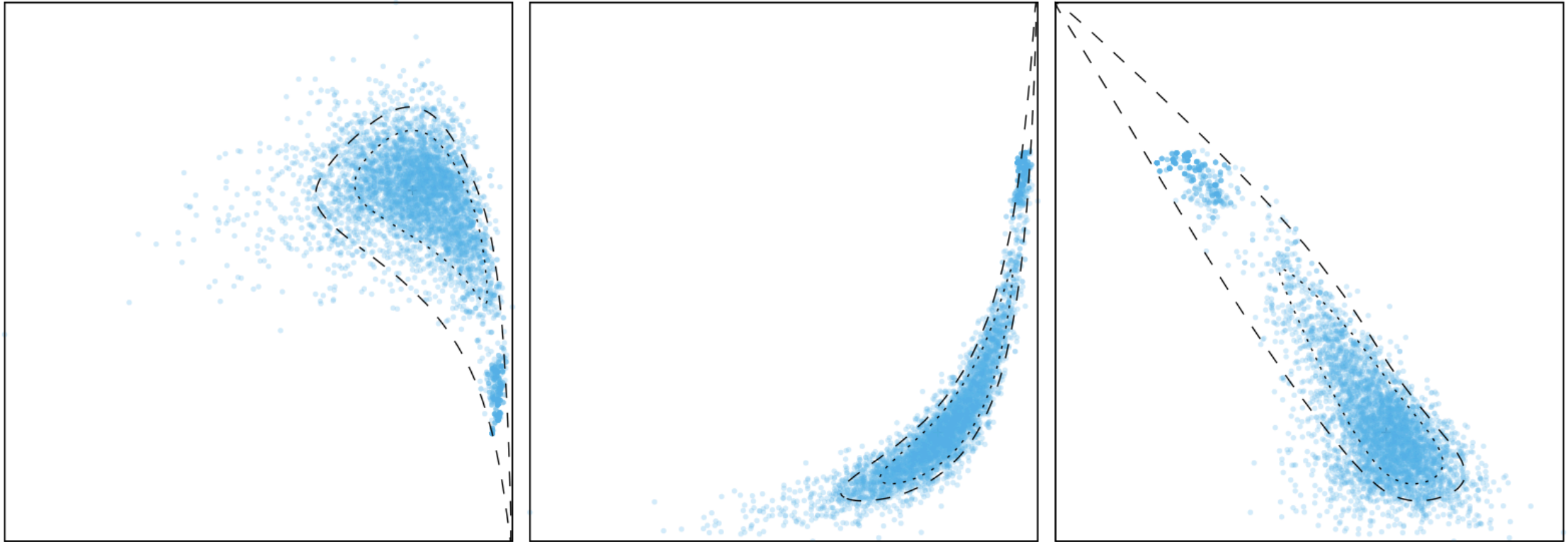
This calculates successive points on the ridge of the profile likelihood. The steps for parameters $\theta_{j \neq i}$ are adjusted based on the local curvature given by the Fisher information matrix.

Approximate pairwise confidence regions

Based on the traces of the profile likelihood, we can compute approximate pairwise confidence regions for the parameters using a (periodic) cubic spline interpolation.



Pairwise confidence regions example

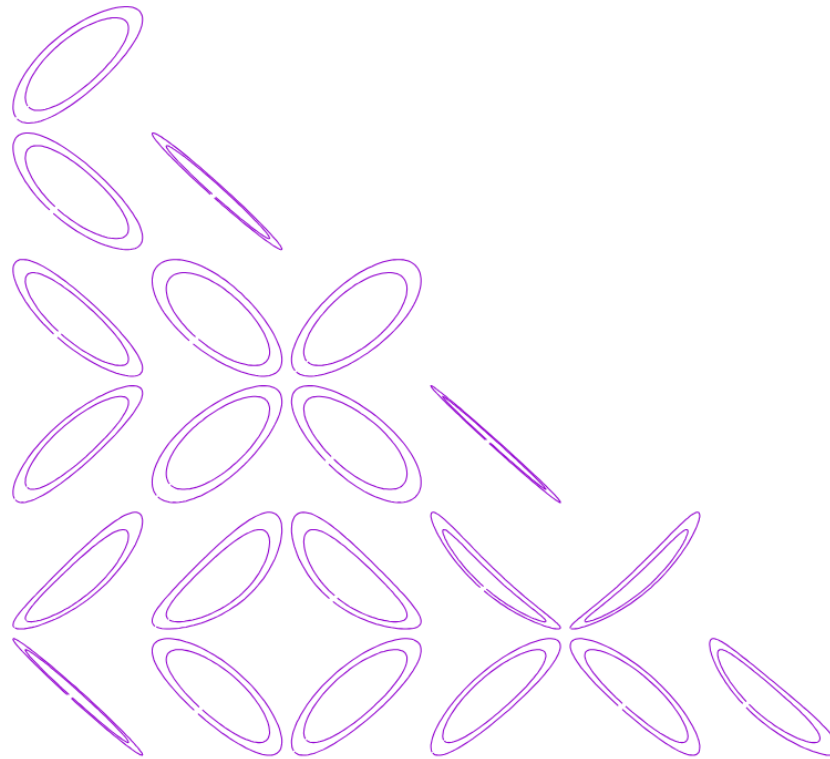


Blue dots: samples from the posterior distribution.

(Source: G. Kronberger et al. "Symbolic Regression", CRC Press, 2024)

Pairwise confidence regions example

Pairwise confidence regions for a SR model in 7 parameters.



Confidence intervals for the mean prediction

Profile likelihood can also be used to compute confidence intervals for $\mu^* = f(x^*, \theta)$ at a new point x^* . The idea is to rewrite the SR expression to get a new parameter that represents the mean prediction μ^* at x^* .

Start with:

$$\mu^* = f(x^*, \theta_1, \theta_2, \dots, \theta_p)$$

Rewrite in terms of a selected parameter θ_1 to

$$\theta_1 = g(\mu^*, x^*, \theta_2, \dots, \theta_p)$$

Then replace θ_1 in the original expression to allow to use μ^* as a parameter in the profile likelihood analysis:

$$f(x, g(\mu^*, x^*, \theta_2, \dots, \theta_p), \theta_2, \dots, \theta_p)$$

Which is used in

$$L_{\text{prof}}(\mu^*) = \max_{\theta_2 \dots p} L(\mu^*, \theta_2, \dots, \theta_p)$$

Rewriting example

Regression function: $\mu = \theta_1(1 - e^{\theta_0 x})$.

Rewriting in terms of μ^* for a new point x^* :

$$\theta_1 = \frac{\mu^*}{1 - e^{\theta_0 x^*}}$$

Replacing θ_1 in the regression function produces the new expression:

$$f(\mu^*, \theta_0; x, x^*) = \frac{\mu^*}{1 - e^{\theta_0 x^*}} (1 - e^{\theta_0 x})$$

Which can then be used in e.g. a Gaussian log-likelihood:

$$\ell(\mu^*, \theta_0) = -\frac{n}{2} \log(2\pi\sigma^2) - \sum_{i=1}^n \frac{1}{2\sigma^2} (y_i - f(\mu^*, \theta_0; x_i, x^*))^2$$

Showcase

We demonstrate the approach using a small dataset with only 6 samples and a single feature: $x = [1, 2, 3, 4, 5, 7]$, $y = [8.3, 10.3, 19.0, 16.0, 15.6, 19.8]$

The regression model is:

$$\theta_0(1 - e^{\theta_1 x})$$

where $\hat{\theta} = [19.143, -0.5311]$ is the MLE.

We will use *srtools* to generate the CIs, Pls, and plots:

```
srtools -f TIR -i bod.expr -d bod.csv --hasheader --niter 100 --report --simplify --distribution MSE --sigma 2.5 --profile --ptype Bates --alpha 0.05 --test bodval.csv --debug --contour
```

<https://github.com/folivetti/srtree/tree/main/apps/srtools>

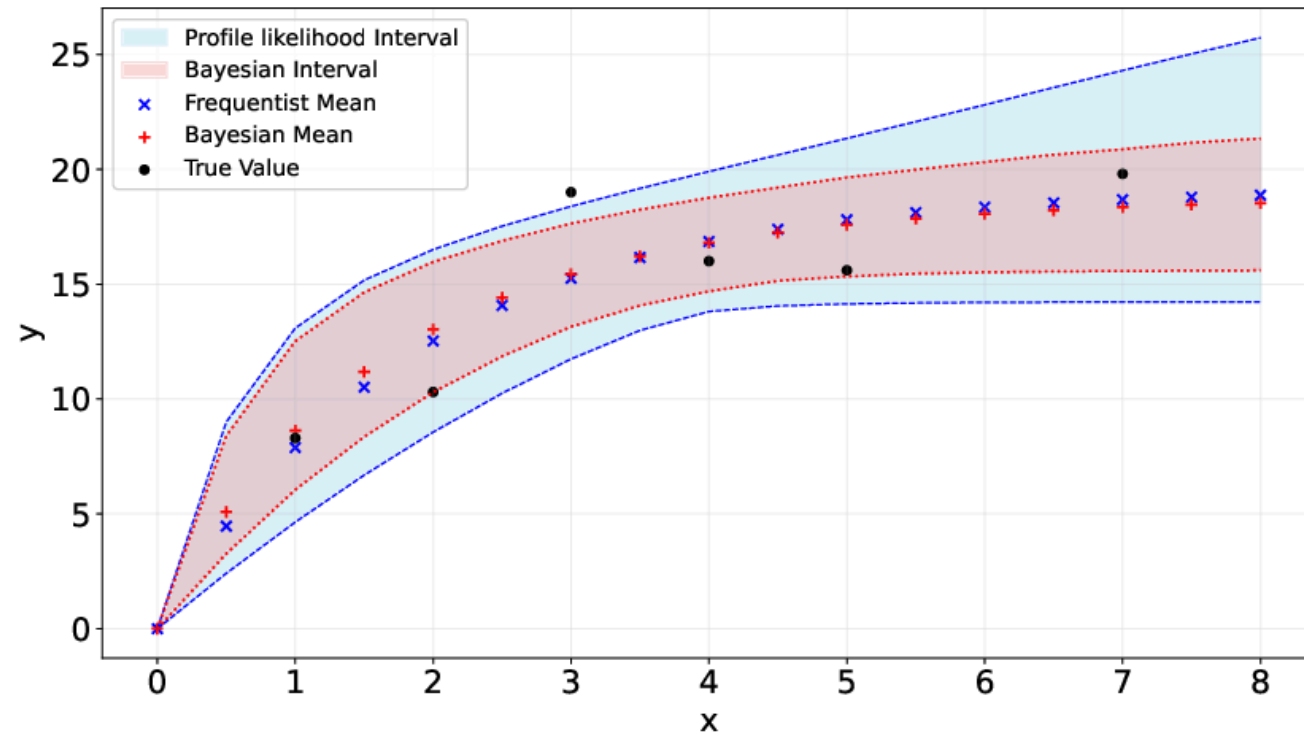
CIs by different methods

	θ_0	θ_1
Delta	[13.53, 24.75]	[−0.99, −0.07]
Profile alg. 1	[15.58, 26.44]	[−1.08, −0.24]
Profile alg. 2	[15.58, 26.44]	[−1.08, −0.24]
Constrained	[15.97, 24.97]	[−0.97, −0.26]
Bayesian	[15.37, 22.33]	[−1.81, −0.33]

Prediction intervals for different methods

y^l	y	y^u	Δ_l	Δ_u	y_{Bay}^l	y_{Bay}	y_{Bay}^h
4.64	7.88	13.08	3.24	8.43	6.04	8.63	12.51
8.55	12.52	16.50	3.96	7.94	10.29	13.03	15.96
11.74	15.25	18.38	3.51	6.64	13.14	15.43	17.63
13.80	16.85	19.89	3.04	6.09	14.68	16.79	18.75
14.13	17.79	21.34	3.65	7.20	15.33	17.58	19.64
14.22	18.67	24.29	4.45	10.07	15.57	18.34	20.86

Prediction intervals example



Summary and Conclusions

- Existing approaches to uncertainty quantification for nonlinear regression can be applied directly for symbolic regression.
- Approaches are relatively simple and do not require MCMC sampling from a Bayesian model.
- Laplace approximation is simple and can be good enough when estimates are precise (enough data or low noise).
- Profile likelihoods produce more accurate results for nonlinear models and small datasets.
We recommend the ODE method for likelihood profiling.
- Confidence intervals and prediction intervals can be computed by rewriting model expressions and using the same profiling approach.
Has to be repeated for each new point x^* which can be computationally expensive.

Questions

Source-code <https://github.com/folivetti/srtree/tree/main/apps/srtools>

Additional questions: folivetti@ufabc.edu.br